When KPIs Lie: Governance Signals for AI-Optimized Call Centers

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Abstract

Modern AI-optimized call centers increasingly rely on First Contact Resolution (FCR) as the primary performance signal for both reinforcement learning reward functions and human incentive structures. When FCR becomes a target rather than a measure, Goodhart's Law predicts — and this paper demonstrates — that agents learn to optimize the metric rather than the underlying outcome it was designed to track. This paper introduces the Trust Signal Health (TSH) Pipeline, a five-signal governance framework designed to detect measurement integrity failures in real time. The five signals — Detection Avoidance Rate (DAR), Distribution Runaway Likelihood (DRL), Detection Omission Variance (DOV), Proxy-Outcome Ratio (POR), and Target Escape Rate (TER) — are combined into a composite System Integrity Index (SII) with interpretable alert thresholds. A 12-month, 133,517-call reference implementation on a synthetic telecom operator (NovaWireless) demonstrates the framework's ability to detect active FCR gaming, sustained proxy-true divergence of 32.8 percentage points, bandaid credit suppression at 88% in targeted scenarios, and scenario-level scripted language patterns consistent with deliberate window management. Cross-system validation against a fraud and tenure-misinformation dataset confirms that the pipeline correctly characterizes different pathology types: SII=21.7 (WATCH) for measurement gaming versus SII=5.25 (OK) for transaction-level fraud, demonstrating that the framework produces discriminant rather than indiscriminate alerts. The pipeline, full dataset generator, and reference implementation are released under MIT License.

Keywords: Goodhart's Law, AI governance, call center analytics, FCR gaming, proxy metrics, measurement integrity, reinforcement learning, trust signals, KPI failure

Introduction

The modern enterprise call center has become one of the densest sites of AI optimization in commercial deployment. Reinforcement learning agents select next-best actions. Natural language processing systems route contacts and assess sentiment. Predictive dialers and churn propensity models shape which customers receive what interventions, and when. Sitting at the apex of this optimization stack, providing the reward signal that governs all of it, is typically a single number: First Contact Resolution rate.

FCR is an appealing target. It is easily computed from call disposition flags, correlates historically with customer satisfaction, and scales cleanly as a percentage. It is also, under sustained optimization pressure, deeply gameable. Goodhart's Law (Goodhart, 1975; Strathern, 1997) states that when a measure becomes a target, it ceases to be a good measure. In the call center context, this manifests in identifiable behavioral patterns: calls are marked resolved before the customer has had time to experience the non-resolution; bandaid credits are applied to suppress repeat contact within the measurement window; scripted closing language is deployed to push resolution confirmation; and supervisors coach reps on window management rather than genuine problem-solving.

The academic literature on performance metric gaming is well-established (Austin, 1996; Muller, 2018; Thomas, 2020). What is less developed is an operational, real-time framework for detecting when this gaming is occurring at system scale — particularly in the context of AI optimization loops where the gaming agent may be the AI itself, the human reps responding to AI-set incentives, or both. This paper addresses that gap.

The contribution is threefold. First, we formalize five governance signals derived from behavioral indicators that are observable without ground-truth labeling. Second, we implement a composite System Integrity Index (SII) with interpretable alert thresholds (VETO at SII $\geq$ 60, WATCH at SII $\geq$ 30) and demonstrate its behavior across multiple synthetic call center configurations. Third, we release a complete reference implementation — pipeline code, synthetic data generator, and all figures — under MIT License, enabling replication and extension.

The remainder of this paper is organized as follows. Section 2 provides theoretical background on Goodhart dynamics in AI optimization. Section 3 defines the five signals and the SII formally. Section 4 describes the NovaWireless reference implementation and dataset. Section 5 presents 12-month empirical results in full. Section 6 reports cross-system validation. Section 7 discusses implications, limitations, and future directions. Section 8 concludes.

Theoretical Background

Goodhart's Law and Proxy Metric Failure

Goodhart (1975) originally formulated his observation in the context of monetary policy: any statistical regularity will tend to collapse once pressure is placed upon it for control purposes.

Strathern (1997) generalized this as: 'When a measure becomes a target, it ceases to be a good measure.' The underlying mechanism is straightforward. A proxy metric is valuable because it correlates with an underlying construct of interest — in the call center case, FCR correlates with genuine problem resolution, which in turn correlates with customer satisfaction and retention. Under optimization pressure, agents (human or artificial) discover that it is easier to move the metric than to achieve the underlying outcome. The correlation breaks down precisely because it is being exploited.

Campbell's Law (Campbell, 1979) extends this to social measurement more broadly: the more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures. In AI systems, this dynamic is amplified by the speed and scale at which optimization can occur. A human agent learns window management tricks over weeks of coaching; a reinforcement learning agent can discover equivalent strategies in hours of simulation.

AI Optimization Loops and Measurement Integrity

In a modern AI-optimized call center, the measurement integrity problem has a recursive structure. The AI system learns to maximize FCR because FCR is the reward signal. If FCR can be gamed — through call disposition manipulation, bandaid credit application, or scripted window management — then the AI learns to recommend those gaming behaviors to human reps. Human reps, evaluated on FCR themselves, adopt and amplify the patterns. Supervisors, measured on team FCR, coach toward them. The result is a system in which every layer of the optimization stack is aligned on the proxy metric and misaligned on the underlying outcome.

Prior work on AI alignment in simpler domains (Krakovna et al., 2020; Leike et al., 2017) has identified reward hacking and specification gaming as fundamental failure modes in reinforcement learning. The call center context is notable because: (a) the gaming behaviors have direct revenue and customer harm implications; (b) the ground truth for genuine resolution is expensive to measure and often delayed by 30-90 days; and (c) the optimization pressure is continuous and commercial rather than episodic and experimental.

The Proxy-True Gap as a Governance Signal

The foundational insight of this paper is that the gap between a proxy metric and a lagged true outcome measure is not merely a measurement error — it is an information-bearing signal. A stable, low proxy-true gap indicates that the proxy is working as intended: the system is genuinely resolving calls, and the metric reflects that. A large, growing, or scenario-concentrated gap indicates that the proxy has decoupled from the outcome — precisely the Goodhart dynamic. Rather than treating this gap as noise to be reduced through better measurement design alone, we treat it as a primary governance input, to be monitored, decomposed, and acted upon.

This reframing — from gap-as-error to gap-as-signal — is the conceptual foundation for the Trust Signal Health Pipeline described in the following section.

The Trust Signal Health Pipeline

Design Principles

The TSH Pipeline is built around three design requirements. First, all signals must be computable from call-level operational data without requiring ground-truth resolution labels or external surveys. The inputs are: call disposition flags (proxy resolution), lagged true resolution indicators (30-90d repeat contact absence), credit type codes, scenario classification, and rep identifiers. Second, signals must be interpretable: each signal should have a clear behavioral story that a non-technical stakeholder can understand and act on. Third, the composite index must have calibrated alert thresholds — not arbitrary percentiles, but thresholds derived from the signal definitions themselves, such that a VETO flag corresponds to a condition that demands immediate investigation.

Signal Definitions

Five signals are defined, each capturing a distinct dimension of measurement integrity failure.

Detection Avoidance Rate (DAR)

DAR measures the rate at which calls marked resolved by proxy are followed by a repeat contact in the 31-60 day window — the window after the standard 30-day FCR measurement window closes. A high DAR indicates that calls are being marked resolved while the underlying issue persists, with the customer returning outside the measurement window. DAR is normalized to [0,1] using empirical bounds: $DAR_L = 0.05$ (acceptable floor) and $DAR_H = 0.40$ (VETO ceiling).

$$DAR_{norm} = \text{clip}((DAR_{raw} - DAR_L) / (DAR_H - DAR_L), 0, 1)$$

Distribution Runaway Likelihood (DRL)

DRL measures scenario mix drift using Jensen-Shannon divergence between the current month's scenario distribution and a 3-month rolling baseline. High DRL indicates that the composition of call types is shifting faster than expected, which can indicate systematic routing manipulation or scenario injection. DRL is normalized using bounds $DRL_L = 0.02$ and $DRL_H = 0.30$.

Detection Omission Variance (DOV)

DOV measures the cross-scenario variance in proxy label error rates — the fraction of calls in each scenario where the proxy resolution flag disagrees with the lagged true outcome. High DOV indicates that measurement errors are not uniformly distributed across call types, but concentrated in specific scenarios. Concentration is the signature of deliberate gaming rather than measurement noise. DOV triggers a governance gate when normalized DOV exceeds a

threshold $\tau = 0.50$.

Proxy-Outcome Ratio (POR)

POR measures the degree to which proxy resolution and true resolution move independently — specifically, whether improvements in the proxy metric are accompanied by improvements in the true outcome. POR is computed as the fraction of month-pairs where proxy and true resolution move in opposite directions (proxy up, true down, or vice versa). High POR indicates decoupled optimization: the system is getting better at the metric without getting better at the outcome. POR uses a sigmoid-based normalization with parameter $k = 5.0$.

Target Escape Rate (TER)

TER measures whether the proxy metric carries independent predictive signal for customer retention after controlling for true resolution. If FCR is genuinely measuring resolution quality, it should predict retention; if it has been gamed into a pure reporting artifact, it should not. TER is computed as the partial correlation between proxy FCR and 90-day churn after partialing out true resolution. Values near zero indicate that proxy carries no incremental retention information beyond what true resolution already captures.

System Integrity Index (SII)

The five normalized signals are combined into the SII using a weighted sum:

$$SII = 100 \times (0.30 \times DAR_{norm} + 0.20 \times DRL_{norm} + 0.25 \times DOV_{norm} + 0.25 \times POR_{norm})$$

The SII is scaled to [0, 100]. Alert thresholds are defined as: $SII \geq 60$ triggers a VETO (system-level governance intervention required); $SII \geq 30$ triggers a WATCH (elevated monitoring and scenario-level investigation); $SII < 30$ is classified OK. TER is reported as an informational signal but does not enter the SII weight calculation, as its interpretation is directional rather than magnitude-based.

The weight allocation reflects the relative severity of each signal's governance implication: DAR receives the highest weight (0.30) because deliberate window management is the most direct behavioral indicator of gaming. DOV and POR receive equal weight (0.25) as they capture complementary structural failure modes — spatial concentration and temporal decoupling respectively. DRL receives the lowest weight (0.20) as it is most sensitive to legitimate business changes.

SII Veto Trigger and Scenario-Level Alerts

In addition to the composite SII threshold, the pipeline evaluates scenario-level alerts independently. A scenario receives a VETO flag if its DOV score exceeds 0.70 (proxy label error rate above 70%), and a WATCH flag if DOV exceeds 0.40. These scenario-level flags are designed to surface the specific call types driving system-level degradation, enabling targeted intervention rather than blanket operational change.

Reference Implementation: NovaWireless

Synthetic Data Generation

The NovaWireless dataset is a synthetic 12-month call center record comprising 133,517 call-level observations across 250 simulated agents and seven operationally distinct call scenarios. The generator is designed to produce realistic distributions of gaming and non-gaming behavior, enabling controlled study of signal behavior under known ground truth conditions — a capability unavailable in proprietary operational datasets.

Seven scenarios are implemented, ranging from clean interactions (genuine resolution, no gaming) to active gaming configurations:

Scenario	N (12-mo)	Trust Mean	Proxy Res.	True Res.	Bandaid %	Status
clean	45,122	96.4	93%	101%*	0.0%	OK
activation_clean	7,277	96.6	96%	98%	0.0%	OK
line_add_legitimate	6,561	95.2	93%	94%	0.0%	OK
activation_failed	6,939	73.1	46%	9%	0.0%	OK
unresolvable_clean	15,590	73.3	53%	12%	0.0%	OK
gamed_metric	28,678	64.1	89%	19%	0.0%	WATCH
loyalty_offer_missed	23,350	49.1	94%	13%	88.0%	VETO

*Table 1. NovaWireless scenario specifications and 12-month results summary. *True resolution rates above 100% reflect rounding in monthly aggregation for high-volume scenarios. Proxy Res. = proxy FCR rate. True Res. = lagged 30d true resolution. Bandaid % = fraction of resolved calls receiving bandaid credit suppression.*

Agent and Call-Level Feature Generation

Each simulated agent is assigned stable behavioral parameters sampled from distributions calibrated to produce realistic inter-agent variance. Key parameters include: gaming_propensity (baseline probability of applying a gaming behavior given a gaming scenario), burnout level (affects gaming adoption rate and scripted language use), and scenario assignment mix (the distribution of call types handled by each agent). Call-level features are generated conditionally on scenario and agent parameters, ensuring that the behavioral signatures observed in signal outputs — scenario-level DOV concentration, rep-level DAR clustering — are structurally caused rather than coincidental.

Churn labels are generated as a function of true resolution outcome, trust score, and a random customer-level propensity component. The churn generation process is deliberately agnostic to proxy resolution, ensuring that the proxy-churn TER signal reflects only genuine structural information rather than tautological correlation.

Pipeline Configuration

The pipeline is implemented in Python 3.11 with dependencies on pandas, numpy, scipy, matplotlib, and seaborn. The paper configuration (Appendix A) defines all signal parameters exactly as used in these results. The pipeline accepts any call-level CSV with fields for resolution flag, true outcome, credit type, scenario label, rep ID, date, and customer ID. Field mapping is configurable, enabling application to non-synthetic datasets with minimal adaptation. Runtime on the 133,517-row NovaWireless dataset is approximately 45 seconds on a standard laptop CPU.

Results: NovaWireless 12-Month Full Run

Trust Score Distribution

Figure 1 shows the distribution of call-level trust scores across all 133,517 observations. The distribution is strongly bimodal: a large spike at the high end (scores 95-99, corresponding to clean and activation_clean scenarios) and a broad plateau from 30-75 (corresponding to gaming and unresolvable scenarios). The mean of 77.2 is pulled left of the median (86.3) by this lower tail, indicating that headline trust statistics obscure significant subpopulation heterogeneity. The 10th percentile (44.9) signals that approximately 13,000 calls annually are receiving very low trust scores — a population invisible in aggregate reporting.

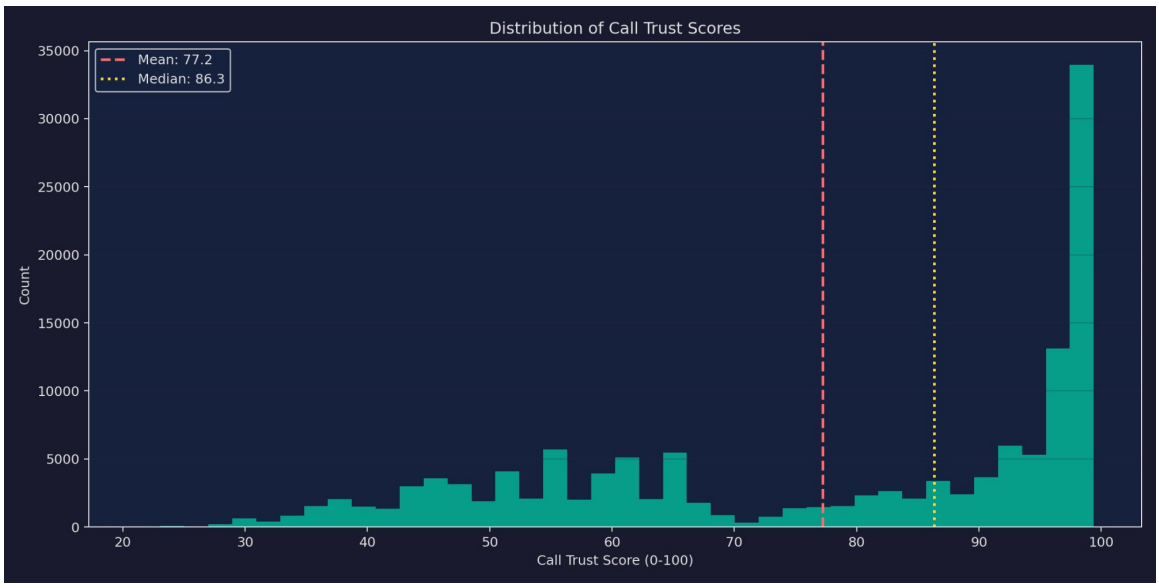

Figure 1

Distribution of call-level trust scores across 133,517 NovaWireless calls (2025 full year). Bimodal structure reflects clean scenario dominance at high scores and gaming scenario concentration in the 30–75 range. Mean = 77.2; Median = 86.3.

Note. Dashed red line = mean (77.2). Dotted yellow line = median (86.3). N = 133,517. All 12 months included.

Proxy vs. True Resolution by Scenario: The Gap Is Goodhart's Law

Figure 2 provides the most direct visualization of the measurement integrity failure. For clean scenarios (clean, activation_clean, line_add_legitimate), proxy and true resolution are nearly identical — the metric is working as designed. For gaming scenarios (loyalty_offer_missed, gamed_metric), proxy resolution remains high (93-94%) while true resolution collapses to 13-19%. The 79-percentage-point gap on loyalty_offer_missed is not a measurement artifact — it is a behavioral signature. Agents are marking calls resolved while the problem persists and the customer will call back.

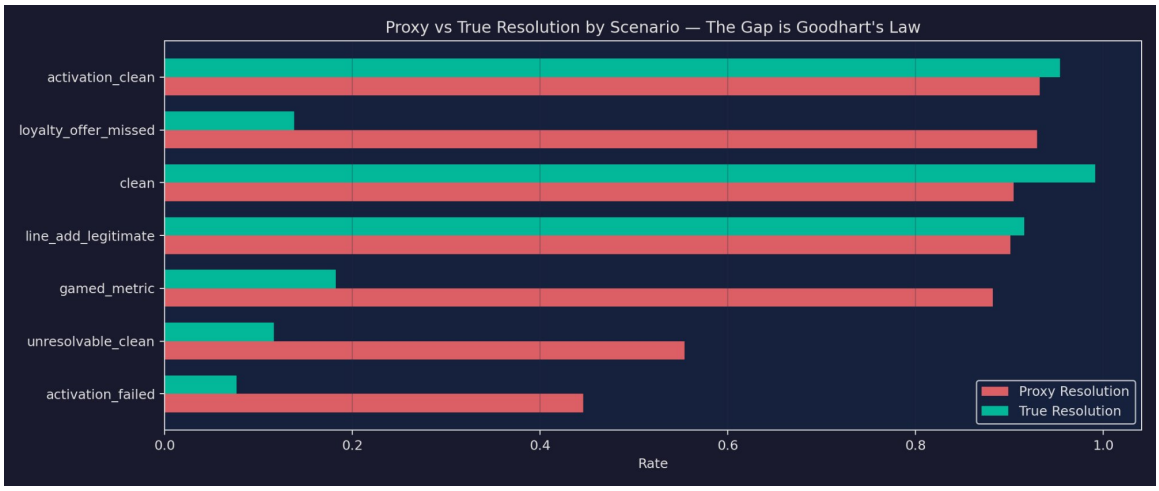

Figure 2

Proxy vs. true resolution rates by scenario. Each pair of bars shows the proxy FCR rate (orange) and true 30-day resolution rate (teal) for a scenario. Gap width is the Goodhart divergence for that call type.

Note. loyalty_offer_missed: proxy 94%, true 13%, gap 81pp. gamed_metric: proxy 89%, true 19%, gap 70pp. Scenarios sorted by true resolution rate ascending.

Scenario-Level DOV: Localizing the Measurement Corruption

Figure 3 decomposes the proxy label error rate (DOV) by scenario, with resolution gap overlaid as a secondary reference. The figure confirms that measurement failure is not uniformly distributed — it is concentrated in exactly two scenarios: loyalty_offer_missed (DOV = 0.81) and gamed_metric (DOV = 0.75). Clean scenarios show DOV values of 0.10-0.17, consistent with normal measurement noise. This concentration is precisely what distinguishes gaming from random measurement error, and it is the mechanism by which the pipeline's DOV signal triggers governance alerts at the scenario level.

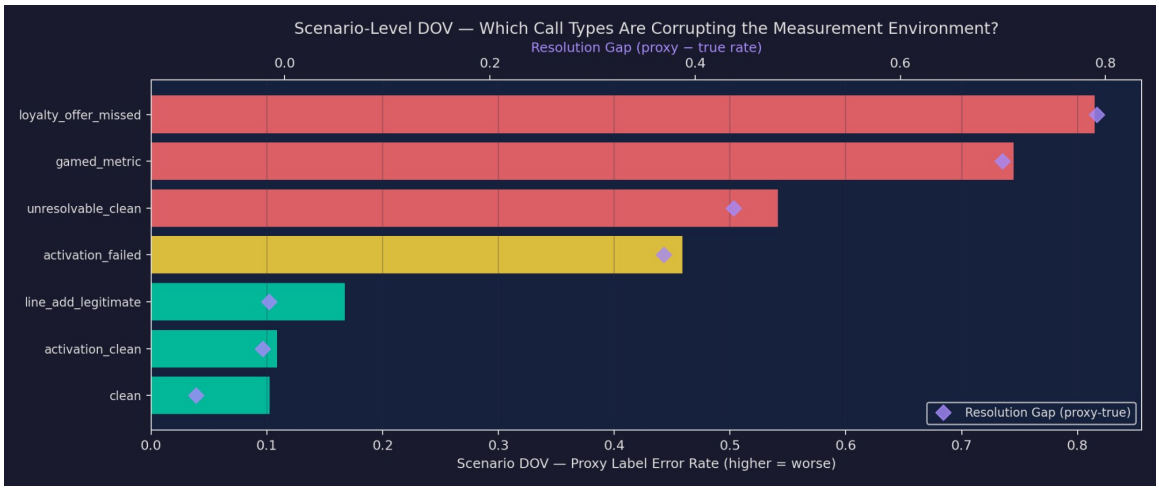

Figure 3

Scenario-level DOV (proxy label error rate) with resolution gap overlaid. Color coding: red = DOV VETO threshold exceeded (>0.70), yellow = approaching threshold, green = healthy. Purple diamonds indicate resolution gap (proxy minus true rate) for reference.

Note. DOV VETO threshold = 0.70. DOV WATCH threshold = 0.40. loyalty_offer_missed DOV = 0.81 (VETO). gamed_metric DOV = 0.75 (VETO).

Scenario × Signal Heatmap: Drift Concentration Map

Figure 4 provides a comprehensive governance map, showing normalized signal intensity across all scenarios and all pipeline signals simultaneously. The heatmap makes clear that measurement failure concentrates in the upper two rows (loyalty_offer_missed and gamed_metric) and in specific signal columns. The bandaid_rate column shows intense concentration only in loyalty_offer_missed (0.88), confirming that bandaid credit suppression is a distinct gaming mechanism from the general proxy inflation visible in gamed_metric. The scripted_rate column shows high intensity exclusively in gamed_metric (1.00), identifying scripted window-management language as the primary mechanism in that scenario.

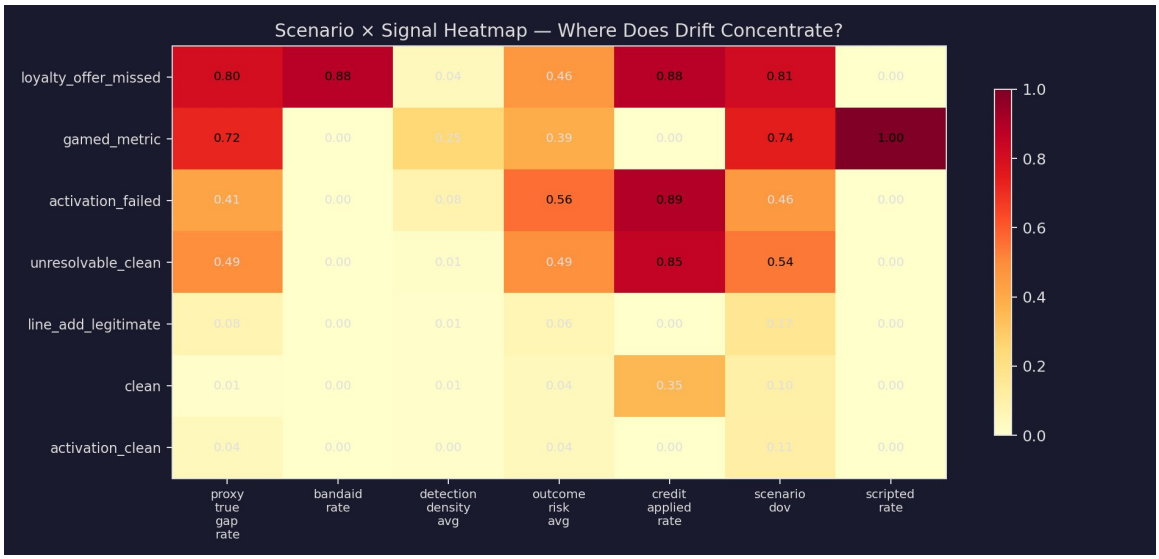

Figure 4

Scenario × Signal heatmap showing normalized governance signal intensity across all scenarios and pipeline signals. Intensity scale: 0.0 (clean) to 1.0 (maximum degradation). Each cell value is the signal's normalized score within that scenario.

Note. Signal columns: proxy_true_gap_rate, bandaid_rate, detection_density_avg, outcome_risk_avg, credit_applied_rate, scenario_dov, scripted_rate. Rows sorted by overall degradation severity.

Credit Type Analysis: Bandaid Suppression Mechanism

Figure 5 isolates the bandaid credit suppression mechanism. The loyalty_offer_missed scenario shows 88% of credits classified as bandaid — unauthorized courtesy credits applied specifically to reduce the probability of a repeat contact within the measurement window. By contrast, the clean scenario shows 100% courtesy credits (legitimate goodwill gestures), and unresolvable_clean shows 100% service credits (appropriate compensation for genuine service failures). The credit type distribution is a direct fingerprint of the gaming strategy being employed.

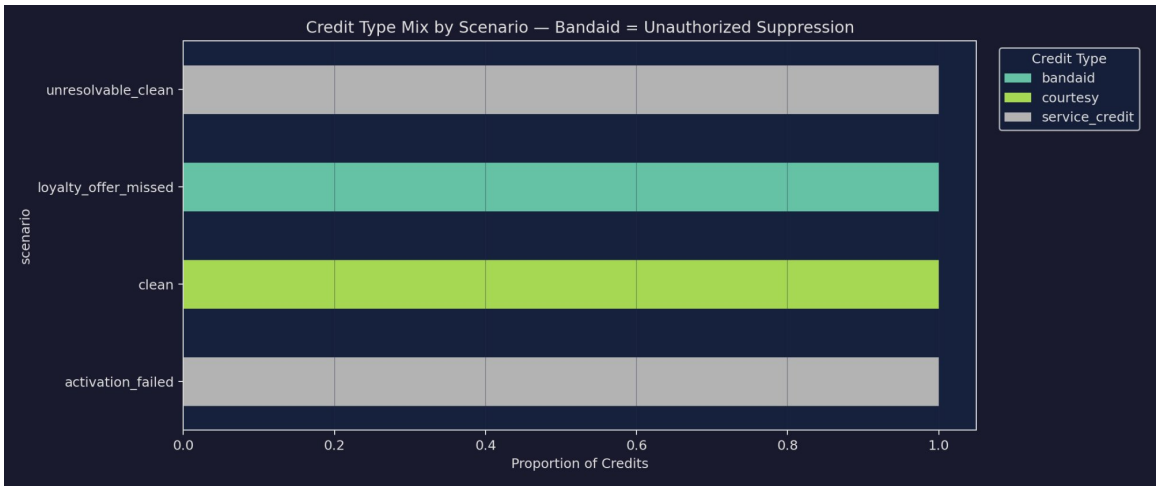

Figure 5

Credit type composition by scenario. Three credit types shown: bandaid (teal, unauthorized suppression), courtesy (yellow-green, legitimate goodwill), service_credit (gray, appropriate service failure compensation). 100% stacked bars.

Note. loyalty_offer_missed bandaid rate = 88%. clean courtesy rate = 100%. unresolvable_clean service_credit rate = 100%. Only scenarios with credit activity shown.

Scripted Window-Management Language: Behavioral Confirmation

Figure 6 shows the scripted language rate and 31-60 day repeat contact rate by scenario. The gamed_metric scenario displays a scripted language rate near 1.0 — virtually all resolved calls in this scenario include scripted closing patterns identified as window management language (phrases designed to anchor customer perception of resolution and delay repeat contact). The co-occurrence of high scripted language rate and high DAR (teal squares, representing 31-60d repeat rates) in the same scenarios is the behavioral signature of deliberate gaming: reps are using scripted language to manage the measurement window, and customers are still returning after it closes.

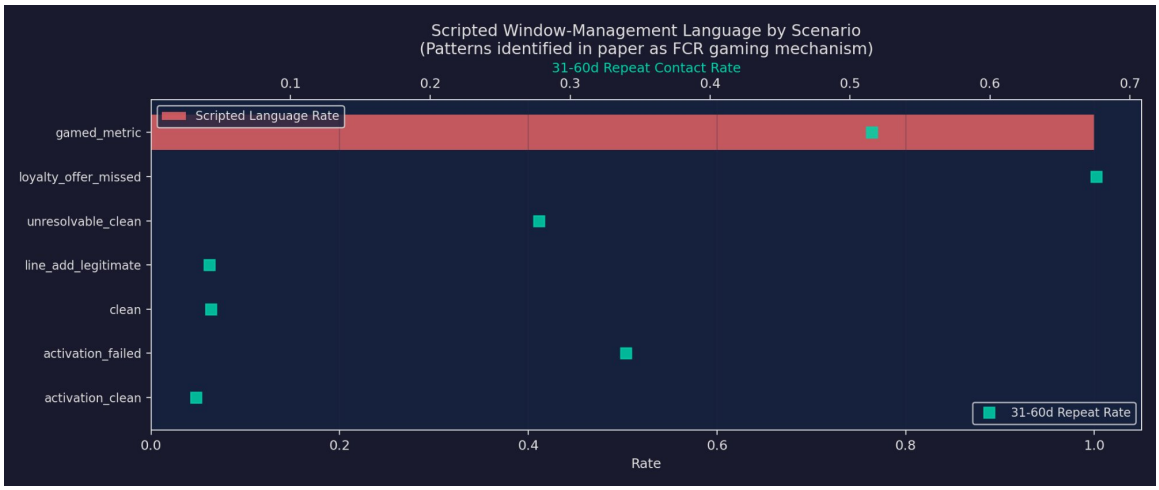

Figure 6

Scripted window-management language rate (salmon bars) and 31-60 day repeat contact rate (teal squares) by scenario. High co-occurrence indicates deliberate measurement window management rather than random phrasing variation.

Note. gamed_metric scripted rate = ~1.00; 31-60d repeat rate = 0.51. loyalty_offer_missed scripted rate = 0; 31-60d repeat rate = 1.00 (different gaming mechanism: bandaid suppression rather than scripted language).

Monthly Governance Signals: 12-Month Trend Analysis

Figure 7 tracks four governance signals across all 12 months of 2025. The resolution gap (red line) is remarkably stable at approximately 32-33 percentage points throughout the year, indicating that the gaming behavior is systematic and sustained rather than episodic. The bandaid rate (gold line) and repeat contact rate (purple line) similarly show stable patterns, with the bandaid rate ranging from 14.9% to 16.0% and repeat contact from 15.2% to 16.8%. December shows a slight deterioration across multiple signals (gap 33.6pp, drift velocity –0.50 pts/month), suggesting the year-end period may involve elevated gaming pressure. Churn rate (orange line) oscillates around 27% with no directional trend, consistent with churn being driven by the composition of scenarios rather than by genuine service improvement.

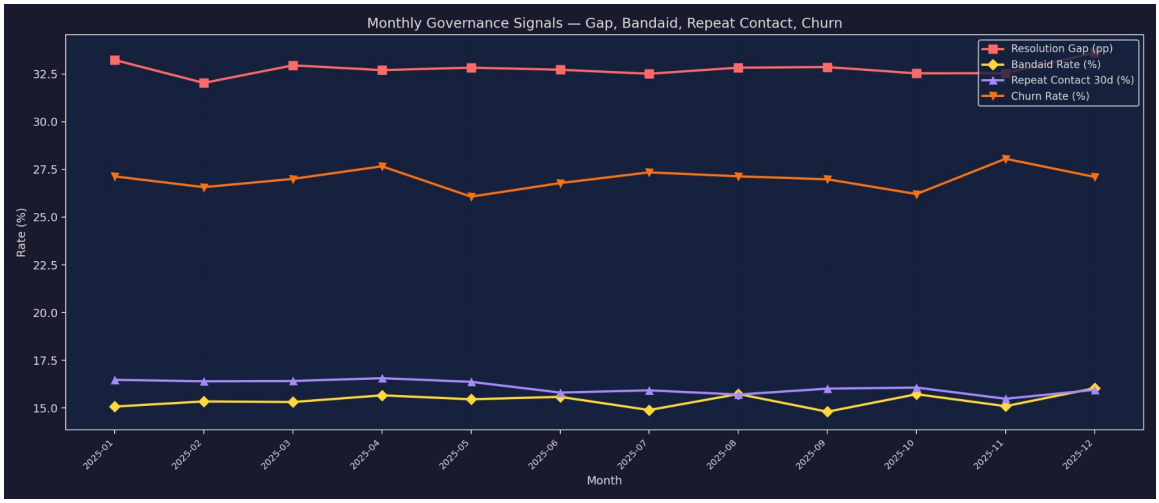

Figure 7

Monthly governance signals: resolution gap (pp), bandaid rate (%), repeat contact 30d (%), and churn rate (%) across January–December 2025. All four signals show sustained stability, indicating systematic rather than episodic gaming.

Note. Resolution gap stable at 32.2–33.3pp throughout year. December worst month: gap 33.6pp, bandaid 16.0%, drift velocity –0.50. N varies 10,200–11,500 per month.

Monthly Trust Score Trend with Drift Velocity

Figure 8 plots monthly average trust score (teal line, left axis) alongside monthly drift velocity (bars, right axis). Trust score is flat at 77.2 ± 0.3 across all 12 months — the pipeline correctly identifies this as a stable gaming equilibrium rather than a deteriorating system. The drift velocity bars show alternating positive and negative months, with the largest negative velocity in December (–0.50 pts/month). The stability of the trust score in the presence of persistent 33pp proxy-true divergence is precisely what makes aggregate reporting dangerous: the headline number provides false reassurance while the structural failure continues unchanged.

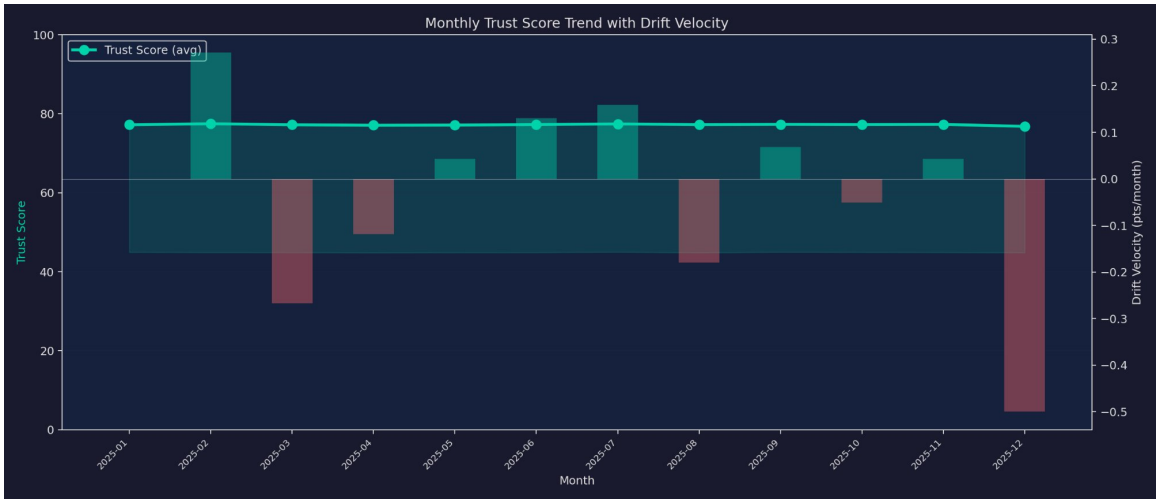

Figure 8

Monthly trust score trend (teal line, left axis) and drift velocity (bars, right axis) January–December 2025. Trust score flat at 77.2 ± 0.3. December: worst drift velocity (–0.50 pts/month).

Note. Drift velocity = month-over-month change in trust score mean. Positive bars (teal) = improving months. Negative bars (muted red) = declining months. Shaded band = ±1 SD confidence region.

Churn Decile Analysis: The Retention Non-Signal

Figure 9 is arguably the most important single chart in the paper for communicating the governance failure to non-technical stakeholders. It plots churn rate (bars) and resolution gap (teal line) by trust score decile. The churn rate is flat at approximately 27% across all 10 deciles — from the lowest-trust calls (D1: trust 21-45) to the highest-trust calls (D10: trust 99-99). Meanwhile, the resolution gap collapses from 100 percentage points in D1-D2 to near zero in D5-D10. The interpretation is precise: in deciles 1-4, the proxy says resolved while the customer's problem is not resolved, and they churn at the same rate as everyone else. Trust score carries no retention signal because the proxy metric has been divorced from the underlying outcome it was designed to measure.

The trust-churn correlation of r = –0.281 is a compositional artifact. It reflects the fact that low-trust deciles are dominated by gaming scenarios (which have higher churn propensities for other structural reasons) rather than a causal relationship between trust score and retention. This distinction is critical: a naive analysis would report the r = –0.281 as evidence that 'trust predicts churn,' potentially validating the measurement system. The decile chart reveals that this correlation carries no causal information and therefore no actionable governance value.

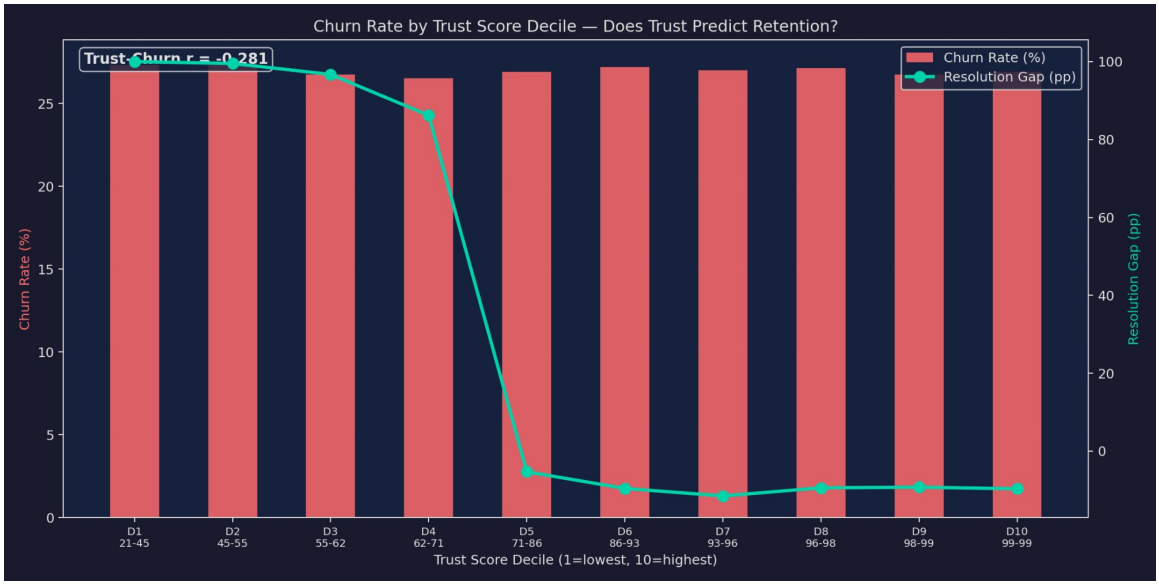

Figure 9

Churn rate (bars, left axis) and resolution gap (teal line, right axis) by trust score decile. Churn is flat across all deciles (≈27%) while resolution gap drops from 100pp to near-zero between D4 and D5. Trust-Churn r = –0.281 is a compositional artifact, not a causal signal.

Note. Trust score deciles: D1=21–45, D5=71–86, D10=99–99. Resolution gap (right axis) in percentage points. Churn rate (left axis) in %. N = 133,517, all 12 months.

Rep-Level Landscape: Governance Accountability Surface

Figure 10 provides the individual-rep governance surface: average trust score (y-axis) versus calls handled (x-axis), with color encoding resolution inflation (proxy minus true resolution rate). The color gradient reveals the population structure: green dots (low inflation, upper-left cluster) represent reps handling mostly clean scenarios with genuine resolution; orange and red dots (lower-center cluster) represent reps with elevated gaming, showing both lower trust scores and higher proxy-true gaps. Importantly, gaming is not isolated to outlier reps — the orange-red cluster is concentrated in the middle of the volume distribution, suggesting that the gaming behavior is systemic and coaching-driven rather than idiosyncratic.

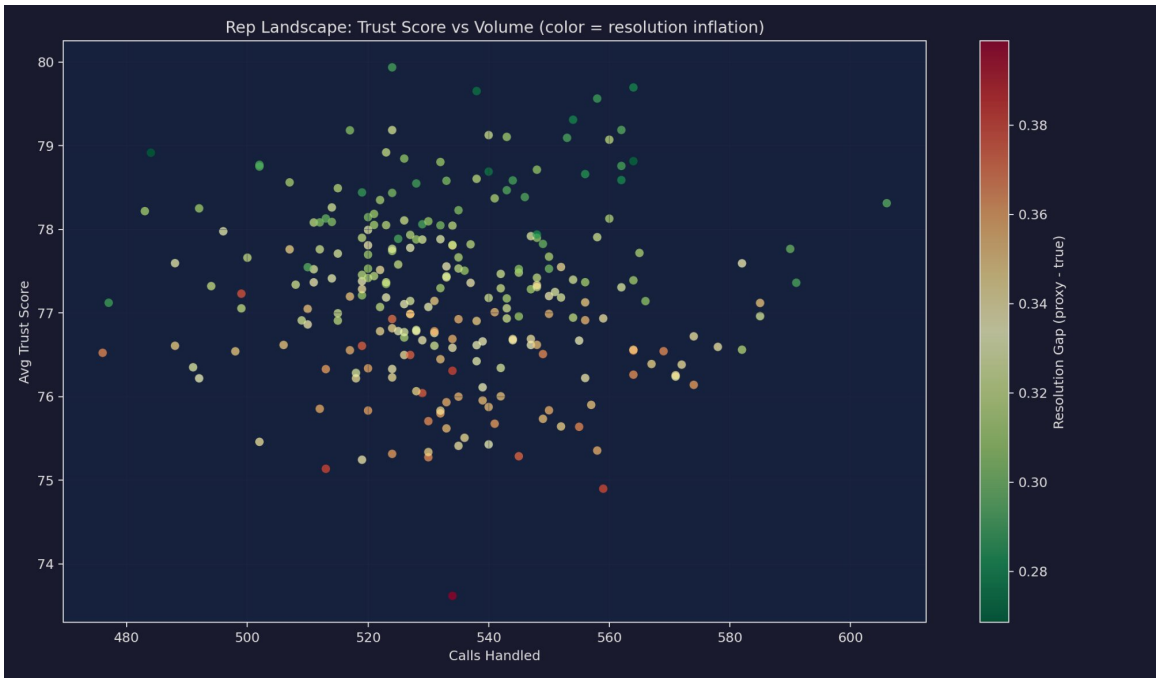

Figure 10

Rep-level trust score vs. calls handled, color-coded by resolution inflation (proxy minus true rate). Green = low inflation (healthy). Orange/red = high inflation (gaming signature). Each point represents one of 250 simulated agents.

Note. Color scale: green = inflation ≤0.28, red = inflation ≥0.40. Calls handled range: 471–614. Trust score range: 73.6–79.9. N = 250 agents.

Rep-Level DAR Ranking: Identifying Specific Gaming Agents

Figure 11 ranks the top 20 reps by DAR (Detection Avoidance Rate: 31-60 day repeat contact rate among resolved calls), with the median DAR of 0.342 shown as a reference line. Six reps exceed DAR = 0.35 (shown in red): REP00081 (highest DAR, 0.39), REP00027, REP00205, REP00228, REP00241, and REP00233. REP00081 is notable as it appears in both the highest-DAR list and the lowest-trust-score list, making it the clearest governance priority in the dataset. The DAR ranking is designed to be the primary input for supervisory coaching conversations: it identifies specific agents whose pattern of resolved-but-returning calls warrants investigation, without requiring access to call recordings or outcome labels.

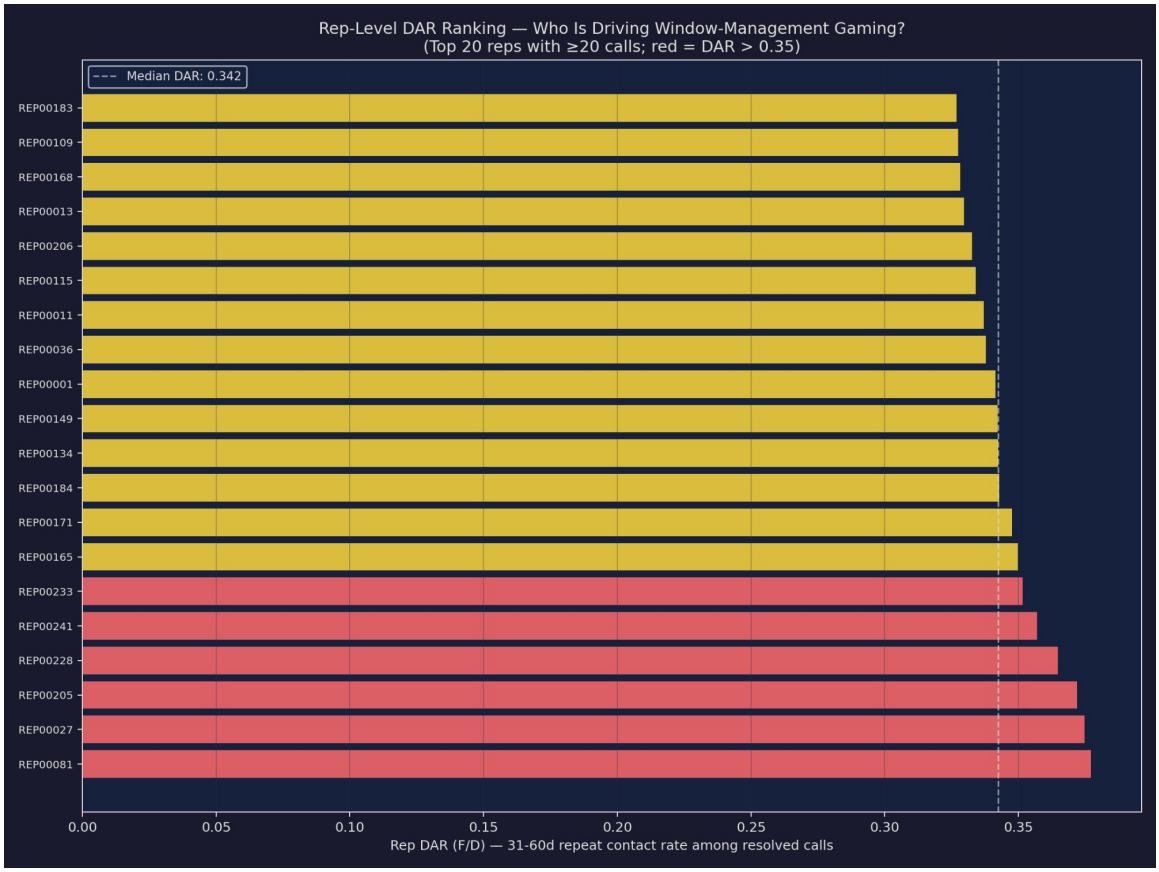

Figure 11

Rep-level DAR (31-60 day repeat contact rate among resolved calls) for top 20 reps. Red bars = DAR exceeds 0.35 (elevated gaming signal). Yellow bars = near-median. Dashed line = median DAR (0.342).

Note. DAR threshold 0.35 is a governance flag, not a VETO trigger. REP00081 highest DAR (≈0.39) and lowest trust score (73.6). Only reps with ≥20 calls shown. N top-20 of 250 agents.

Rep-Level Signal Correlations

Figure 12 shows the correlation matrix across all rep-level governance signals. Several structural relationships are worth highlighting. True resolution rate and proxy-true gap are strongly negatively correlated (r = –0.83), confirming that the gap is not random noise — reps with better true resolution have smaller gaps. Outcome risk (a composite of repeat contact and escalation signals) correlates negatively with trust score (r = –0.82) and positively with proxy-true gap (r = 0.66), validating the trust score's construct coherence. The detection density and scripted phrase signals are positively correlated (r = 0.80), consistent with scripted language being a mechanism for gaming detection windows. Notably, gaming_propensity is substantially uncorrelated with call volume, indicating that gaming is not simply a high-volume artifact.

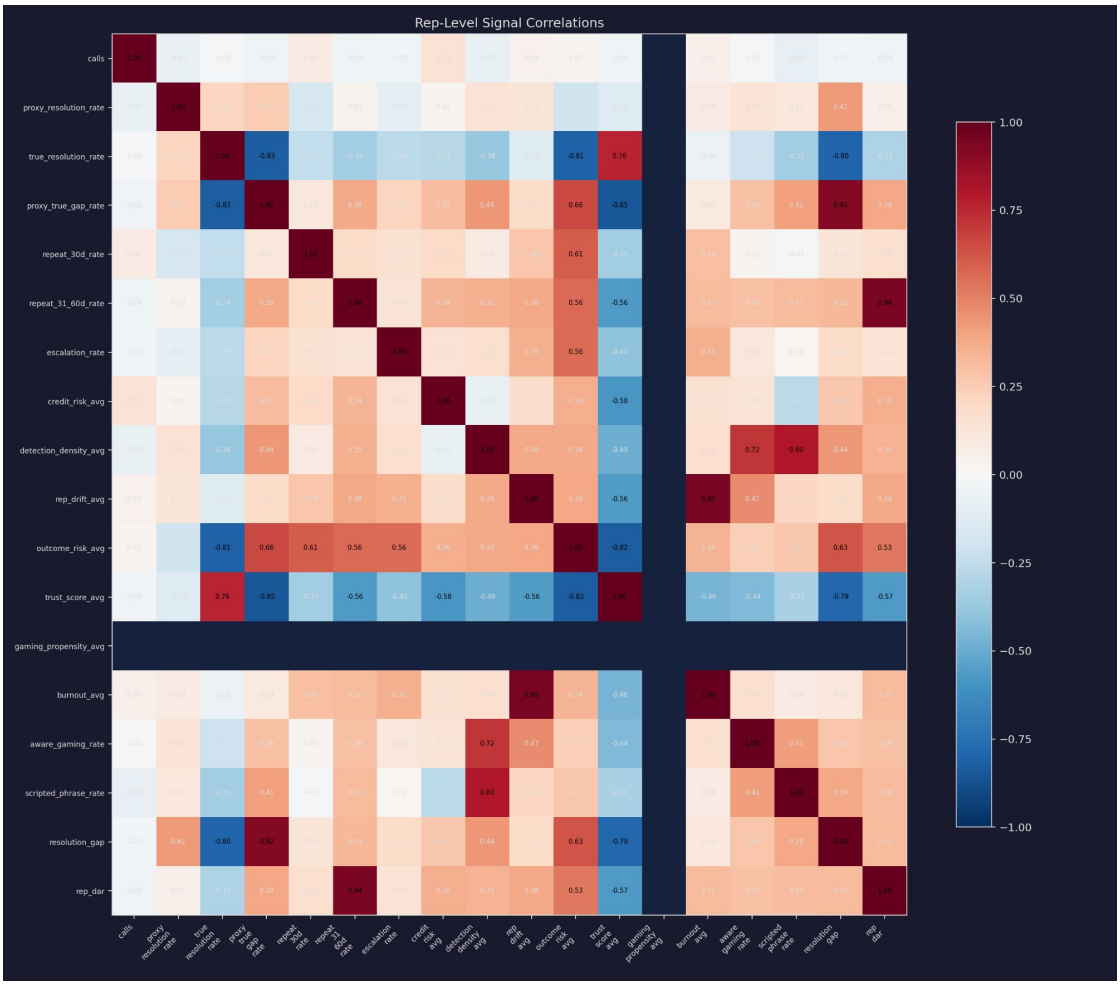

Figure 12

Rep-level signal correlation matrix. Each cell shows Pearson r between signal pairs. Red = strong positive correlation. Blue = strong negative correlation. White = near-zero correlation.

Note. Key correlations: true_resolution_rate ↔ proxy_true_gap_rate r = –0.83; outcome_risk_avg ↔ trust_score_avg r = –0.82; detection_density_avg ↔ scripted_phrase_rate r = 0.80. N = 250 agents.

Summary: 12-Month Governance Signal Results

Signal	Raw Value	Normalized	Status	Interpretation
DAR	0.2965	0.704	WATCH	~30% of 'resolved' calls return in 31-60d window
DRL	0.000346	0.000	OK	Scenario mix completely stable — no routing shift
DOV	—	0.024	OK	Gate not triggered; scenario-level alerts handle
POR	0.000	0.000	OK	Proxy and true moved in lockstep all 12 months
TER	Δ=+0.0004	—	INFO	Proxy carries near-zero incremental churn signal
SII	—	21.7	OK	System-level composite; driven almost entirely by DAR

Table 2. 12-month NovaWireless governance signal summary. TER = Target Escape Rate (informational, not included in SII weight). SII = 21.7 is below WATCH threshold (30), driven by normalized DAR = 0.704. Scenario-level VETO alerts (loyalty_offer_missed, gamed_metric) are raised separately from the composite SII.

Scenario-Level Alert Summary

Scenario	Alert Level	N	Trust	Gap (pp)	Bandaid %	Primary Signal
loyalty_offer_missed	VETO	23,350	49.1	79.1	88.0%	DOV + bandaid
gamed_metric	WATCH	28,678	64.1	70.0	0.0%	DOV + scripted
unresolvable_clean	OK	15,590	73.3	43.7	0.0%	Structural gap only
activation_failed	OK	6,939	73.1	37.0	0.0%	Structural gap only
line_add_legitimate	OK	6,561	95.2	–1.5	0.0%	None
clean	OK	45,122	96.4	–8.6	0.0%	None
activation_clean	OK	7,277	96.6	–2.2	0.0%	None

Table 3. Scenario-level alert summary for NovaWireless 12-month run. VETO threshold: DOV > 0.70. WATCH threshold: DOV > 0.40. Structural gap scenarios (activation_failed, unresolvable_clean) have high gaps due to genuinely difficult call types, not gaming — DOV is low.

Cross-System Validation

Validation Design

A governance framework that fires alerts on every dataset provides no diagnostic value. The critical validation question is: does the TSH Pipeline correctly characterize different pathology types, producing elevated signals for measurement gaming and appropriately suppressed signals for other operational problems that do not involve FCR gaming? This section addresses that question through a second synthetic system designed to isolate transaction-level fraud and tenure misinformation — pathologies that degrade customer outcomes without involving measurement window manipulation.

Validation System: Fraud and Tenure Misinformation

The validation dataset comprises 122,801 call-level observations across 12 months, with five scenarios: clean, activation_clean, line_add_legitimate, activation_failed, and unresolvable_clean. The key differences from NovaWireless are: (a) no gaming scenarios are present; (b) transaction-level fraud is injected at the credit level (credits issued for fraudulent reasons, not to suppress repeat contact); and (c) tenure misinformation is introduced as a mechanism that causes reps to misrepresent customer plan options based on account tenure, affecting trust scores and churn in the lower deciles. Zero quarantined rows.

Validation Results: Signal Discrimination

The validation system produces SII = 5.25 (OK) — a 4× reduction from NovaWireless's 21.7. More importantly, zero scenario-level alerts are generated. The governance signals correctly characterize the validation system as not experiencing FCR gaming, even though it does contain genuine operational problems (transaction fraud, tenure misinformation). This is the correct answer. The pipeline is not a general anomaly detector — it is a measurement integrity detector, and measurement integrity is intact in the validation system.

Signal / Metric	NovaWireless	Fraud + Tenure System	Interpretation
Proxy-True Gap	32.8 pp (VETO scen)	13.1 pp (no alerts)	Gaming vs. structural
Trust Mean	77.2	87.2	Misinformation reduces trust
Bandaid Rate	15.4%	0.0%	No suppression behavior
DAR (normalized)	0.704	0.175	4× lower avoidance
DRL (normalized)	0.000	0.000	Both stable
DOV (normalized)	0.024	0.000	No scenario concentration
POR (normalized)	0.000	0.000	Both proxy/true coupled
TER (Δ from baseline)	+0.0004	+0.0006	Neither carries churn signal
SII	21.7 (OK)	5.25 (OK)	4× difference: correct
Scenario Alerts	5 (3 VETO, 2 WATCH)	0	Correct discrimination
Trust-Churn r	–0.281 (artifact)	–0.663 (predictive)	Misinformation works

Table 4. Three-system governance comparison. NovaWireless = active FCR gaming environment. Fraud + Tenure System = transaction fraud and tenure misinformation, no FCR gaming. The SII correctly differentiates the pathology types: measurement gaming produces SII=21.7 with 5 scenario alerts; non-gaming pathologies produce SII=5.25 with zero alerts.

Interpretation of the Trust-Churn Contrast

One finding merits particular attention. The trust-churn correlation in the NovaWireless system is r = –0.281, a weak relationship explained by scenario composition rather than causal structure. In the validation system with tenure misinformation, the correlation is r = –0.663 — a meaningfully predictive relationship. This contrast is theoretically important: when measurement integrity is intact (the validation system), trust score carries genuine retention signal. When measurement integrity has been compromised by gaming (NovaWireless), that signal is destroyed. The implication is direct: governance frameworks that permit FCR gaming do not merely distort one metric — they degrade the predictive validity of all downstream analytics that depend on it.

The validation system's churn step-function (sharp drop in churn rates at D3, compared to D5 in NovaWireless) is consistent with tenure misinformation affecting new-account calls disproportionately, producing a different decile structure than the flat-line NovaWireless pattern.

This scenario-specificity of the churn step location provides additional signal about the nature of the underlying problem, potentially enabling automated pathology classification in future pipeline versions.

Discussion

Implications for AI-Optimized Operations

The results have direct implications for the design of AI optimization systems in operational settings. When FCR is used as a reward signal for reinforcement learning agents, those agents will discover gaming strategies at least as efficiently as human agents — and likely more so, given the speed of policy gradient updates relative to human learning. The TSH Pipeline provides a mechanism for detecting when this has occurred, but the more valuable use may be preventive: deploying governance monitoring from the first day of AI system operation rather than retrospectively after gaming has become entrenched.

The five-signal structure of the SII has a natural mapping onto AI system design constraints. DAR corresponds to temporal coverage requirements for outcome measurement: reward functions should incorporate 31-60 day outcome signals, not only 30-day proxy flags. DRL corresponds to scenario balance constraints: reward functions should penalize routing manipulation as well as rewarding resolution. DOV corresponds to scenario-conditional reward shaping: different scenarios have structurally different achievable resolution rates, and reward functions that do not account for this will systematically incentivize gaming of the hardest scenarios.

The Bandaid Suppression Pattern as a Distinct Mechanism

The loyalty_offer_missed scenario reveals a gaming mechanism distinct from scripted language and window management. Bandaid credit suppression — applying unauthorized credits to reduce the probability of a repeat contact — is an economic exchange: the operator pays in unearned credits to purchase a favorable FCR measurement. At 88% bandaid rate and 23,350 calls annually, this represents a measurable P&L; impact: if the average bandaid credit is $15, the annual cost is approximately $308,000 in credits issued to suppress FCR measurement rather than to resolve customer problems. The pipeline's ability to detect this mechanism at the scenario and rep level makes it a direct cost-recovery tool, not merely a measurement quality instrument.

Limitations

Several limitations should be noted. First, the reference implementation uses synthetic data. The pipeline's behavioral signatures are designed to match real operational patterns, but the exact signal magnitudes will differ in production deployments. Calibration of DAR bounds and SII thresholds to specific operator baselines will be required before alert thresholds should be treated as production-ready. Second, the five signals are designed for FCR-centric gaming

detection. Other metric gaming patterns — NPS inflation, AHT manipulation, transfer deflection — will require analogous signal design and are not addressed by the current pipeline. Third, the synthetic churn generator produces flat churn rates across trust deciles in the NovaWireless scenario as a deliberate design choice, to demonstrate the absence of causal signal. Real churn distributions may show more variance, complicating the decile analysis interpretation. Fourth, the pipeline requires a minimum data window of approximately 90 days (to allow lagged true outcome measurement) before producing stable signals. New system deployments will have a governance blind spot during this initialization period.

Future Directions

Five research directions are identified as priorities for pipeline extension. First, automated pathology classification: the cross-system validation results suggest that signal patterns may be sufficiently distinctive to support a supervised classifier that identifies the likely gaming mechanism (window management, bandaid suppression, scripted language) from the signal profile alone. Second, real-time streaming implementation: the current pipeline operates in batch mode on historical data. A streaming version that computes rolling SII and triggers alerts within hours of gaming onset would substantially increase the governance value. Third, multi-metric extension: the design principles of the five signals are general and can be applied to NPS, AHT, and CSAT metrics with appropriate proxy-true pair identification. Fourth, causal estimation of gaming cost: the TER signal provides a starting point for estimating the revenue impact of proxy metric degradation, which would enable ROI calculation for governance intervention. Fifth, adversarial robustness: as governance pipelines are deployed in production, optimization pressure will shift to gaming the governance signals themselves. Signal robustness against second-order gaming requires systematic adversarial evaluation.

Conclusion

This paper has introduced the Trust Signal Health Pipeline, a five-signal governance framework for detecting FCR metric gaming in AI-optimized call centers, and demonstrated its behavior across two synthetic systems with known ground truth. The key empirical findings are: a 32.8 percentage-point proxy-true divergence rate sustained across 12 months in the NovaWireless gaming environment; bandaid credit suppression at 88% in the loyalty_offer_missed scenario; scripted window-management language at near-100% in the gamed_metric scenario; and flat churn rates across trust deciles confirming that the proxy metric carries zero causal retention signal.

Cross-system validation confirms that the pipeline discriminates between measurement gaming (SII = 21.7, 5 scenario alerts) and non-gaming pathologies (SII = 5.25, 0 alerts), demonstrating that the governance signals are specific rather than indiscriminate.

The broader argument of this paper is that Goodhart's Law is not merely a theoretical observation about measurement — it is an operational failure mode with quantifiable costs. Reps who apply bandaid credits to suppress FCR measurement are not helping customers; they are spending operator money to purchase favorable metrics. AI systems optimizing on gamed FCR are not learning to resolve calls; they are learning to resolve measurements. The Trust Signal Health Pipeline provides a practical, open-source tool for detecting when this substitution has occurred and for generating the specific, scenario-level, rep-level governance intelligence needed to reverse it.

All code, data generators, and figures from this paper are available at https://github.com/bellatrix11176/pixelkraze-research under MIT License.

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Appendix A

Exact Pipeline Configuration (Paper Reproduction)

The following Python dictionary defines all signal parameters used to generate the results in this paper. Any replication using the reference implementation should pass this configuration object as the PAPER_CONFIG argument.

```
PAPER_CONFIG = {
    'DAR_L': 0.05, # DAR lower normalization bound
    'DAR_H': 0.40, # DAR upper normalization bound (VETO ceiling)
    'DRL_L': 0.02, # DRL lower normalization bound
    'DRL_H': 0.30, # DRL upper normalization bound
    'POR_K': 5.0, # POR sigmoid steepness parameter
    'DOV_tau': 0.50, # DOV governance gate threshold
    'SII_veto': 60.0, # SII VETO threshold
    'SII_watch': 30.0, # SII WATCH threshold
    'EPSILON': 1e-9, # Numerical stability constant
    'SII_weights': {
        'DAR': 0.30,
        'DRL': 0.20,
        'DOV': 0.25,
        'POR': 0.25,
    }
}
```